

Feature engineering

```
graph TD; FE[Feature engineering] --> FT[Feature transformation]; FE --> FC[Feature construction]; FE --> FS[Feature selection]; FE --> FE2[Feature extraction];
```

Feature transformation

change a column's form

missing values
categorical data
outliers
feature scaling
binning, log, ...

Feature construction

build new columns by hand

combine columns
split a column
group values

Feature selection

keep the useful columns

forward selection
backward elimination

Feature extraction

build new columns by an algorithm

PCA
LDA
t-SNE

both reduce the number of columns